IGCSE 0607 Practice Questions — E5.2

Angle measurement in degrees

Total Questions: 21 — **Total Marks:** 122 Marks

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- **E5.2** Angle measurement in degrees
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

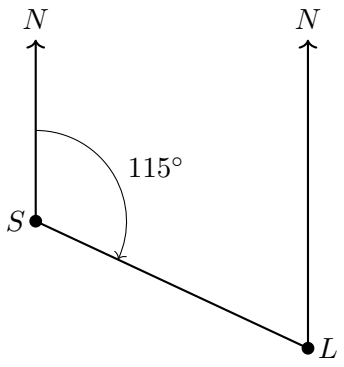
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1 The bearing of point Y from point X is $048°$.

Work out the bearing of point X from point Y.

..... ° **[2]**

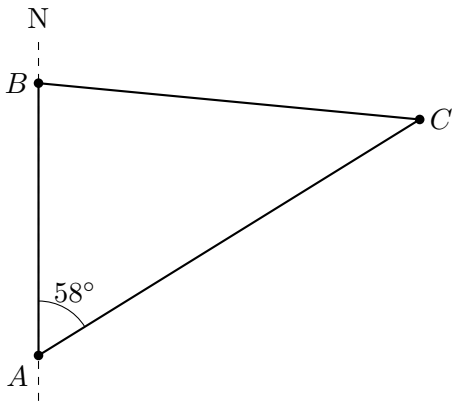
2 The diagram shows the positions of a ship, S, and a lighthouse, L.



The bearing of the lighthouse from the ship is 115°.
Work out the bearing of the ship from the lighthouse.

..... ° **[2]**

3 The diagram shows the positions of three towns, A, B and C.
B is due north of A and the bearing of C from A is 058°.



(a) Find the bearing of A from C.

..... ° **[2]**

(b) Given that angle $ABC = 84°$, find the bearing of C from B.

..... ° **[1]**

4 The positions of three towns, A, B and C, are marked on a map.

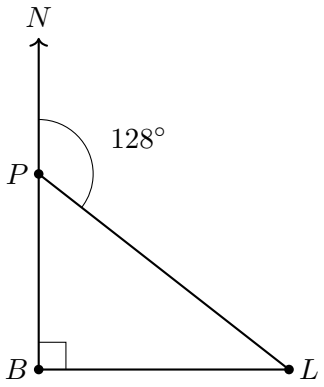
(a) The bearing of town B from town A is 068°.
Work out the bearing of town A from town B.

..... ° **[2]**

(b) Town C is due west of town B.
Write down the bearing of town C from town B.

..... ° **[1]**

5 The bearing of a lighthouse L from a port P is 128°.
A boat B is positioned such that it is due South of P and due West of L.



(a) Find the bearing of P from L.

..... ° **[2]**

(b) Work out the size of angle BPL.

..... ° **[1]**

(c) Work out the size of angle PLB.

..... ° **[2]**

6 The bearing of a lighthouse, L, from a boat, B, is 148°.

(a) Work out the bearing of B from L.

..... ° **[2]**

(b) A port, P, is due north of L and due east of B.

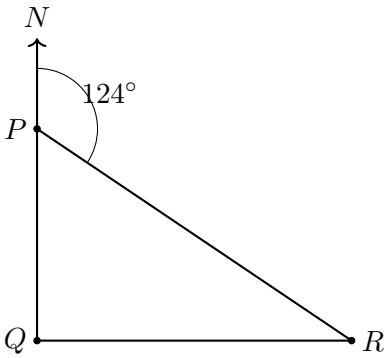
(i) Write down the bearing of B from P.

..... ° **[1]**

(ii) Work out the size of angle LBP.

..... ° **[2]**

7 The diagram shows the positions of three points, P, Q and R.
Point Q is due south of point P.
Point R is due east of point Q.
The bearing of point R from point P is $124°$.



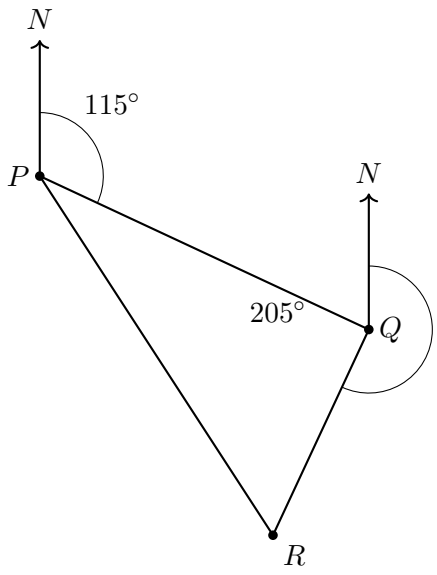
(a) Find the bearing of point P from point R.

..... ° **[2]**

(b) Calculate the angle PRQ.

Angle PRQ = ... ° **[2]**

8 The diagram shows the positions of three points P, Q and R.
The bearing of Q from P is 115°.
The bearing of R from Q is 205°.



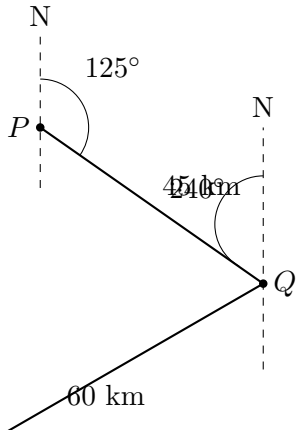
(a) Show that angle $PQR = 90°$.

[2]

(b) The distance PQ is 15 km and the distance QR is 8 km.
Calculate the distance PR.

..... km **[3]**

9 A ship sails from port P on a bearing of 125° for 45 km to point Q.
It then sails from Q on a bearing of 240° for 60 km to point R.



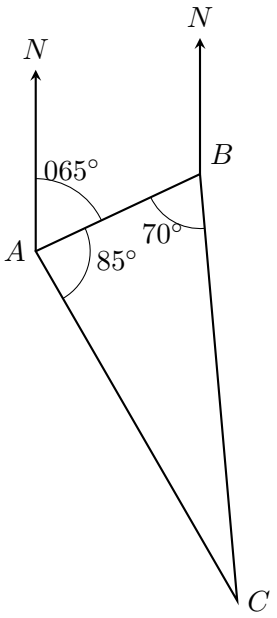
(a) Show that angle $PQR = 65°$.

[2]

(b) Calculate the distance PR.

$PR =$ ..... km **[3]**

10 The diagram shows the positions of three towns, A, B and C.
The bearing of B from A is 065°.
Angle $CAB = 85°$ and angle $ABC = 70°$.



(a) Calculate the bearing of C from A.

..... ° **[1]**

(b) Calculate the bearing of A from B.

..... ° **[2]**

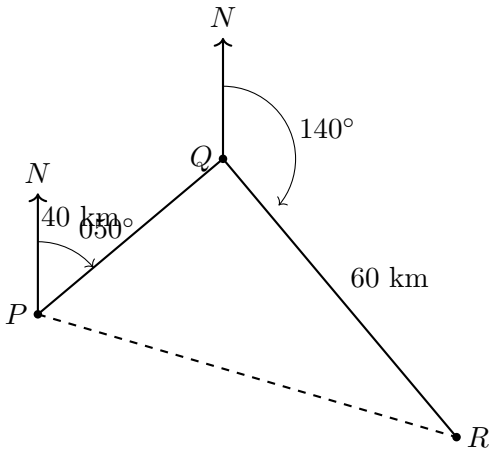
(c) Calculate the bearing of C from B.

..... ° **[2]**

(d) Calculate angle ACB.

..... ° **[1]**

11 A ship sails 40 km from a port P to point Q on a bearing of 050°.
It then sails 60 km from Q to point R on a bearing of 140°.



(a) Show that the angle PQR is 90°.

[2]

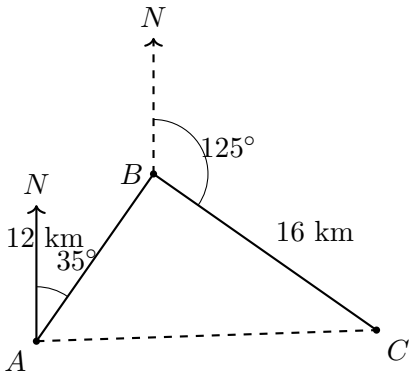
(b) Calculate the distance PR.

..... km **[2]**

(c) Calculate the bearing of R from P.

..... ° **[3]**

12 A ship leaves a port A and travels 12 km on a bearing of 035° to point B.
From B, the ship travels 16 km on a bearing of 125° to point C.



(a) Show that angle $ABC = 90°$.

[2]

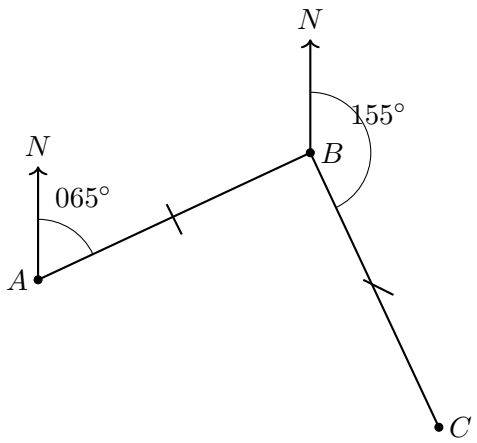
(b) Calculate the direct distance AC.

$AC =$ ..... km **[3]**

(c) Calculate the bearing of C from A.

..... ° **[3]**

13 Point B is on a bearing of 065° from point A.
Point C is on a bearing of 155° from point B.
The distance $AB = BC$.



(a) Show that angle $ABC = 90°$.

[2]

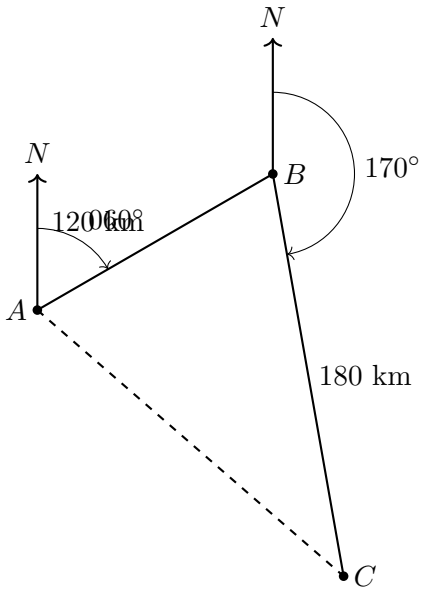
(b) Work out the bearing of A from C.

..... ° **[3]**

(c) Point D is such that the quadrilateral $ABCD$ is a square.
Find the bearing of D from A.

..... ° **[2]**

14 A helicopter flies from base A to base B, a distance of 120 km on a bearing of 060°.
It then flies from base B to base C, a distance of 180 km on a bearing of 170°.



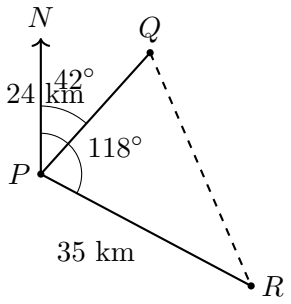
(a) Calculate the distance AC.

..... km **[4]**

(b) Calculate the bearing of C from A.

..... ° **[3]**

15 Three towns are located at points P, Q, and R.
Town Q is 24 km from P on a bearing of 042°.
Town R is 35 km from P on a bearing of 118°.



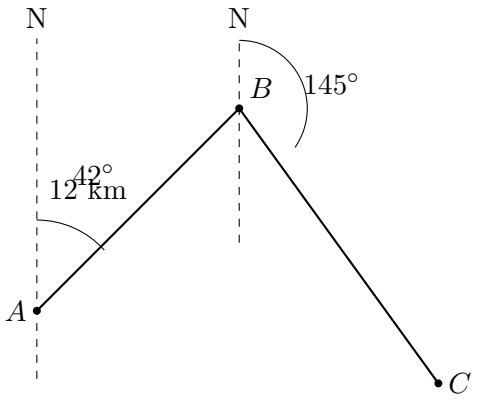
(a) Calculate the distance QR.

$QR =$ ..... km **[3]**

(b) Calculate the bearing of R from Q.

..... ° **[4]**

16 Three lighthouses A, B and C are situated such that B is 12 km from A on a bearing of 042°. C is 18 km from A and the bearing of C from B is 145°.



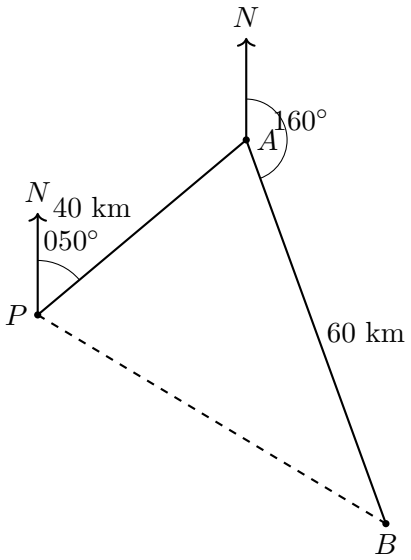
(a) Calculate the bearing of C from A.

..... ° **[5]**

(b) A boat sails from A in a straight line such that it is always equidistant from B and C. Calculate the bearing on which the boat sails.

..... ° **[2]**

17 A ship leaves port P and sails 40 km on a bearing of 050° to reach point A.
It then sails 60 km on a bearing of 160° to reach point B.



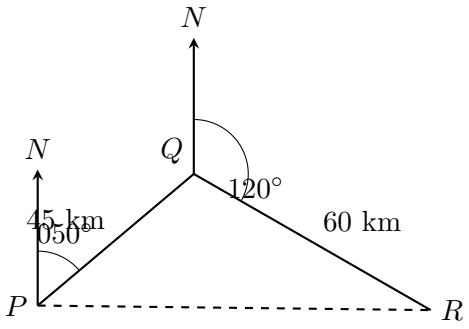
(a) Calculate the distance PB.

..... km **[4]**

(b) Calculate the bearing of B from P.

..... ° **[4]**

18 A ship sails 45 km from Port P on a bearing of 050° to Point Q.
It then sails 60 km on a bearing of 120° to Point R.



(a) Calculate the distance PR.

..... km **[4]**

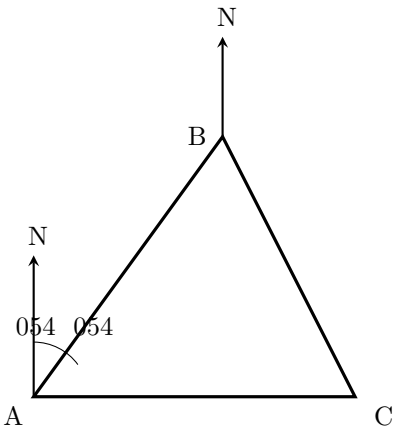
(b) Calculate the bearing of R from P.

..... ° **[3]**

(c) The ship sails directly back from R to P.
Calculate the bearing of P from R.

..... ° **[2]**

19 The diagram shows the positions of three lighthouses,
A,
B, and
C. Lighthouse
C is due east of lighthouse
A, and the bearing of
B from
A is
054
054
.



(a) Write down the three-figure bearing of
C from
A.

.....

[1]

(b) Calculate the three-figure bearing of
A from
B.

.....

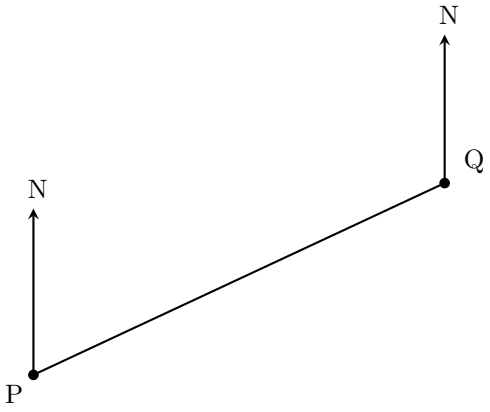
[2]

(c) Given that
=
AB=AC, calculate the three-figure bearing of
C from
B.

.....

[2]

20 The scale drawing shows the positions of two towns,
P and
Q, on a map.
The scale is
1
1 cm represents
5
5 km.



(a) Measure the three-figure bearing of
Q from
P.

.....

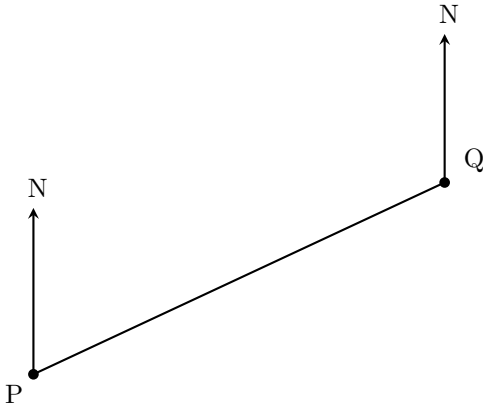
[1]

(b) Work out the actual distance, in kilometres, from town
P to town
Q.

..... km **[2]**

(c) A third town,
R, is
25
25 km from
Q on a bearing of
140
140
.
Using a ruler and protractor, mark and label the position of town

R on the scale drawing below.



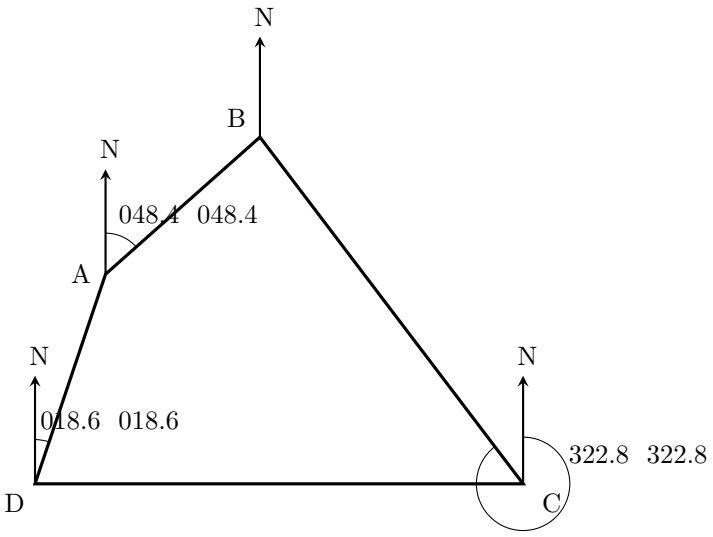
[2]

(d) Without measuring, calculate the three-figure bearing of P from Q.

.....

[2]

21 The diagram shows the positions of four radar stations,
A,
B,
C, and
D, on flat ground.
Station
C is due east of station
D. The bearing of
A from
D is
018.6
018.6
, the bearing of
B from
A is
048.4
048.4
, and the bearing of
B from
C is
322.8
322.8
.



(a) Write down the three-figure bearing of
D from
C.

.....

[1]

(b) Calculate the three-figure bearing of
D from
A.

. . . . . .

[2]

(c) Calculate the size of

(i) angle DAB,

Angle = DAB= . . . . . .

[2]

(ii) angle ABC.

Angle = ABC= . . . . . .

[3]

(d) Calculate the three-figure bearing of C from B.

. . . . . .

[2]